Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (38 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. The answer key is at the back.

Part A · Multiple choice

Choose one answer for each question.

A1 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO1

In all forms of chromatography, separation depends on:

A. different distribution of components between two phases
B. different colours of the components
C. the mass of the column
D. evaporation of the mobile phase

A2 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO1

In a chromatogram, the retention time is used mainly to:

A. help identify a component
B. measure the amount
C. calculate the pressure
D. find the column length

A3 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO1

Separation by molecular size uses:

A. size-exclusion chromatography
B. ion-exchange chromatography
C. adsorption TLC
D. gas–liquid chromatography

A4 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO2

Normal-phase chromatography uses:

A. a polar stationary phase and a less polar mobile phase
B. a nonpolar stationary phase and a polar mobile phase
C. a gas mobile phase
D. a charged resin

A5 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO2

On a silica TLC plate, all spots stay near the origin. The best change is to:

A. use a more polar solvent
B. use a less polar solvent
C. use a longer plate
D. spray with ninhydrin

A6 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO2

Which solvent has the highest polarity index?

A. acetonitrile
B. toluene
C. n-hexane
D. diethyl ether

A7 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO3

R_f values always lie between:

A. 0 and 1
B. 1 and 10
C. −1 and 1
D. 0 and 100

A8 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO3

In paper chromatography, the stationary phase is:

A. water held in the cellulose
B. the solvent
C. silica gel
D. air

A9 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO4

A strong base anion exchanger carries:

A. $-N(CH_3)_3^+$
B. $-SO_3^-$
C. $-COO^-$
D. –OH only

A10 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO4

Which detector is used in GC but not in HPLC?

A. flame ionisation detector
B. UV/Vis detector
C. refractive index detector
D. fluorescence detector

A11 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO5

Residual solvents left from drug manufacture are measured by:

A. headspace GC
B. paper chromatography
C. ion exchange
D. size exclusion

A12 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO5

An internal standard is added in HPLC of blood samples to:

A. correct for losses during sample preparation and injection
B. make the drug fluorescent
C. increase the pressure
D. remove proteins

Part B · Structured questions

Write your answers on paper, then self-mark with the worked solution.

Question B1 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO1

7 marks, C3

(a) Define chromatography and state the role of the stationary and mobile phases. (2 marks)

(b) Explain why two components of a mixture leave a column at different times. (1 mark)

(c) On an HPLC column, t_m = 0.50 min. Peak A: t_r = 3.50 min, w = 0.40 min. Peak B: t_r = 4.10 min, w = 0.50 min. (4 marks)

(i) Calculate the retention factor of each peak.

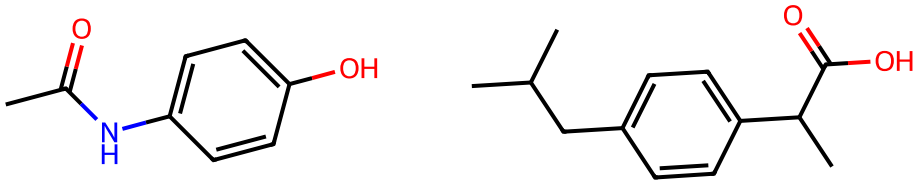
(ii) Calculate the resolution and state whether the peaks are fully separated.

Question B2 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO2

8 marks, C3

(a) Distinguish between normal-phase and reversed-phase chromatography, giving one example of a stationary phase and a mobile phase for each. (2 marks)

(b) Paracetamol (left) and ibuprofen (right) are analysed on a C18 column with water–methanol. Predict which elutes first and explain why. (2 marks)



(c) **(i)** Calculate the polarity index of 70:30 methanol–water (methanol 5.1, water 10.2). (2 marks)

(ii) State the effect on retention times if the mobile phase is changed to 50:50.

(d) Distinguish between isocratic and gradient elution. (2 marks)

Question B3 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO3

7 marks, C3

A pharmacist tests a white powder labelled 'paracetamol'. A paracetamol standard (S) and the test sample (T) are run side by side on a silica gel F_{254} plate.

(a) Describe how the plate is prepared, developed and visualised. (3 marks)

(b) The solvent front moved 8.0 cm. Spot S moved 3.2 cm. Sample T gave two spots, at 3.2 cm and 5.6 cm. Calculate the R_f of each spot. (2 marks)

(c) Interpret the result, and state how you would confirm the identity of the main spot. (2 marks)

Question B4 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO4

8 marks, C2

(a) Describe a cation exchanger and an anion exchanger, naming the functional group on a strong and a weak example of each. (3 marks)

(b) Na^+, Al^{3+} and Ca^{2+} are loaded on a cation-exchange column. State the order in which they are held, from most to least strongly, and explain how they can be washed off. (2 marks)

(c) Name the four main parts of an HPLC instrument after the solvent reservoir, and state the job of any two. (2 marks)

(d) Explain why ethanol in blood can be measured by GC but insulin cannot. (1 mark)

Question B5 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO5

8 marks, C4

A regulator's lab suspects that a 'herbal' slimming capsule contains sibutramine.

(a) Explain how TLC could be used as a quick screen. (2 marks)

(b) The capsule contents are extracted and diluted 1 in 20. By HPLC, the calibration line is: peak area = 1200 + 5400 × C (C in mg/L). The diluted extract gives a peak area of 28 200. Calculate the sibutramine concentration in the diluted extract and in the original extract. (2 marks)

(c) Suggest a method to measure traces of dichloromethane left in a capsule from manufacture, with a reason. (2 marks)

(d) When measuring a drug in a patient's blood by HPLC, explain why (2 marks)

(i) the sample is cleaned up first and

(ii) an internal standard is added.

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Adapted from David Harvey, *Analytical Chemistry 2.1*, chapter 12, and *Instrumental Analysis* 28.6, LibreTexts, CC BY-NC-SA 4.0. TLC questions, drug questions and pharmacy scenarios are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.

PHD125 Pharmaceutical Organic Chemistry · Application of Chromatographic Methods in Pharmaceutical Analysis

Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (38 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO1

In all forms of chromatography, separation depends on:

A. different distribution of components between two phases

B. different colours of the components

C. the mass of the column

D. evaporation of the mobile phase

Harvey 12.1.

A2 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO1

In a chromatogram, the retention time is used mainly to:

A. help identify a component

B. measure the amount

C. calculate the pressure

D. find the column length

Peak area measures amount.

A3 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO1

Separation by molecular size uses:

A. size-exclusion chromatography

B. ion-exchange chromatography

C. adsorption TLC

D. gas–liquid chromatography

Porous gel; large molecules out first.

A4 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO2

Normal-phase chromatography uses:

A. a polar stationary phase and a less polar mobile phase

B. a nonpolar stationary phase and a polar mobile phase

C. a gas mobile phase

D. a charged resin

Silica with, for example, hexane–ethyl acetate.

A5 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO2

On a silica TLC plate, all spots stay near the origin. The best change is to:

A. use a more polar solvent

B. use a less polar solvent

C. use a longer plate

D. spray with ninhydrin

A more polar eluent moves spots further on silica.

A6 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO2

Which solvent has the highest polarity index?

A. acetonitrile

B. toluene

C. n-hexane

D. diethyl ether

5.8, versus 2.4, 0.1 and 2.8.

A7 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO3

R_f values always lie between:

A. 0 and 1

B. 1 and 10

C. −1 and 1

D. 0 and 100

The spot cannot pass the front.

A8 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO3

In paper chromatography, the stationary phase is:

A. water held in the cellulose

B. the solvent

C. silica gel

D. air

Separation is mainly by partition.

A9 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO4

A strong base anion exchanger carries:

A. $-N(CH_3)_3^+$

B. $-SO_3^-$

C. $-COO^-$

D. –OH only

Quaternary amine stays charged.

A10 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO4

Which detector is used in GC but not in HPLC?

A. flame ionisation detector

B. UV/Vis detector

C. refractive index detector

D. fluorescence detector

The FID burns the gas stream.

A11 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO5

Residual solvents left from drug manufacture are measured by:

A. headspace GC

B. paper chromatography

C. ion exchange

D. size exclusion

They are volatile.

A12 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO5

An internal standard is added in HPLC of blood samples to:

A. correct for losses during sample preparation and injection

B. make the drug fluorescent

C. increase the pressure

D. remove proteins

Harvey 12.5 (fluoxetine).

Part B · Structured questions

Question B1 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO1 7 marks, C3

(a) Define chromatography and state the role of the stationary and mobile phases. (2 marks)

(b) Explain why two components of a mixture leave a column at different times. (1 mark)

(c) On an HPLC column, t_m = 0.50 min. Peak A: t_r = 3.50 min, w = 0.40 min. Peak B: t_r = 4.10 min, w = 0.50 min. (4 marks)

- **(i)** Calculate the retention factor of each peak.
- **(ii)** Calculate the resolution and state whether the peaks are fully separated.

(a) Chromatography separates the components of a mixture as a mobile phase (liquid or gas) carries them over a stationary phase (solid, or a liquid film on a solid).

(b) Each component distributes itself between the phases differently: the one held more strongly by the stationary phase moves more slowly.

(c) **(i)** $k_A = (3.50 - 0.50) \div 0.50 =$ **6.0**; $k_B = (4.10 - 0.50) \div 0.50 =$ **7.2**.

(ii) $R = 2(4.10 - 3.50) \div (0.40 + 0.50) = 1.20 \div 0.90 =$ **1.33**. This is below 1.5, so the peaks overlap slightly: not baseline separated.

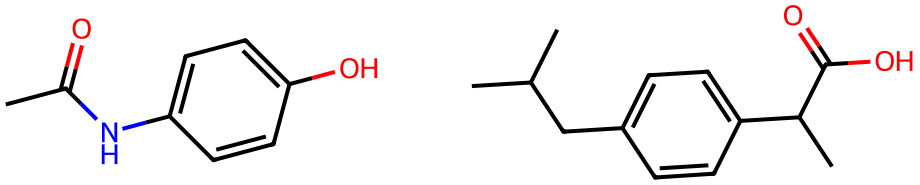
Step	Marking criteria	Marks
1	Definition: separation of mixture components	1
2	Roles of mobile and stationary phases	1
3	Different distribution between phases	1
4	k_A = 6.0	1
5	k_B = 7.2	1
6	R = 1.33	1
7	Below 1.5: not fully separated	1
	Total	**7**

Question B2 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO2

8 marks, C3

(a) Distinguish between normal-phase and reversed-phase chromatography, giving one example of a stationary phase and a mobile phase for each. (2 marks)

(b) Paracetamol (left) and ibuprofen (right) are analysed on a C18 column with water–methanol. Predict which elutes first and explain why. (2 marks)



(c) **(i)** Calculate the polarity index of 70:30 methanol–water (methanol 5.1, water 10.2). (2 marks)

(ii) State the effect on retention times if the mobile phase is changed to 50:50.

(d) Distinguish between isocratic and gradient elution. (2 marks)

(a) Normal phase: polar stationary phase (silica gel) with a less polar mobile phase (hexane–ethyl acetate); less polar compounds elute first. Reversed phase: nonpolar stationary phase (C18 bonded silica) with a polar mobile phase (water–methanol or water–acetonitrile); more polar compounds elute first.

(b) **Paracetamol** elutes first. It has a phenol –OH and an amide group and is more polar; ibuprofen has a large nonpolar hydrocarbon part and is held longer by the nonpolar C18.

(c) **(i)** $P' = 0.70(5.1) + 0.30(10.2) = 3.57 + 3.06 =$ **6.6**.

(ii) 50:50 has $P' = 7.65$, a more polar (weaker) mobile phase in reversed phase, so compounds are held longer: retention times increase.

(d) Isocratic: the mobile phase composition stays the same throughout. Gradient: the composition changes during the run (in reversed phase, from more water to more organic solvent), so late peaks come out sooner and sharper.

Step	Marking criteria	Marks
1	Normal phase described with example	1
2	Reversed phase described with example	1

3	Paracetamol elutes first	1
4	Reason: more polar, less held by C18	1
5	P′ = 6.6	1
6	Retention times increase	1
7	Isocratic defined	1
8	Gradient defined	1
	Total	**8**

Question B3 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO3

7 marks, C3

A pharmacist tests a white powder labelled 'paracetamol'. A paracetamol standard (S) and the test sample (T) are run side by side on a silica gel F_{254} plate.

(a) Describe how the plate is prepared, developed and visualised. (3 marks)

(b) The solvent front moved 8.0 cm. Spot S moved 3.2 cm. Sample T gave two spots, at 3.2 cm and 5.6 cm. Calculate the R_f of each spot. (2 marks)

(c) Interpret the result, and state how you would confirm the identity of the main spot. (2 marks)

(a) Draw a pencil origin line about 1 cm from the bottom and apply small spots of S and T; let them dry. Place the plate in a closed, solvent-saturated tank with the solvent below the origin, and develop until the front is about 1 cm from the top. Remove, mark the front immediately, dry, and view under UV light at 254 nm: spots appear dark on a green background; circle them in pencil.

(b) S: 3.2 ÷ 8.0 = **0.40**. T: 3.2 ÷ 8.0 = **0.40** and 5.6 ÷ 8.0 = **0.70**.

(c) T contains a compound with the same R_f as paracetamol, plus a second substance (an impurity or another drug), so it is not pure paracetamol. Confirm with a co-spot (S + T in one lane gives a single spot at 0.40) or by HPLC retention time or MS.

Step	Marking criteria	Marks
1	Pencil origin, small spots	1
2	Closed saturated tank, solvent below origin; mark the front	1
3	Visualise under UV 254 nm	1
4	R_f(S) = 0.40	1
5	R_f of T spots 0.40 and 0.70	1
6	Contains paracetamol plus another substance	1
7	Confirm by co-spot, HPLC or MS	1
	Total	**7**

Question B4 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO4

8 marks, C2

(a) Describe a cation exchanger and an anion exchanger, naming the functional group on a strong and a weak example of each. (3 marks)

(b) Na^+, Al^{3+} and Ca^{2+} are loaded on a cation-exchange column. State the order in which they are held, from most to least strongly, and explain how they can be washed off. (2 marks)

(c) Name the four main parts of an HPLC instrument after the solvent reservoir, and state the job of any two. (2 marks)

(d) Explain why ethanol in blood can be measured by GC but insulin cannot. (1 mark)

(a) A cation exchanger has fixed negative groups that hold and exchange cations: strong, $-SO_3^-$ (sulfonic acid); weak, $-COO^-$ (carboxylic acid). An anion exchanger has fixed positive groups that exchange anions: strong, $-N(CH_3)_3^+$ (quaternary amine); weak, $-NH_3^+$ (amine). Both are usually polystyrene–divinylbenzene resins.

(b) $Al^{3+} > Ca^{2+} > Na^+$ (higher charge is held more strongly). Elute by raising the concentration of a competing cation in the mobile phase (for example, more HCl or NaCl) or by changing the pH.

(c) Pump (pushes the mobile phase through at high pressure, steady flow); injector (loop, delivers an exact sample volume); column (separates the components); detector (for example UV, produces a signal as each compound leaves).

(d) Ethanol is volatile and stable when heated, so it can be vaporised; insulin is a large, non-volatile protein that would decompose in the hot injector.

Step	Marking criteria	Marks
1	Cation exchanger: negative fixed groups, exchanges cations	1
2	Anion exchanger: positive fixed groups, exchanges anions	1
3	Strong and weak groups named for both	1
4	$Al^{3+} > Ca^{2+} > Na^+$	1
5	Elute with more competing ion or pH change	1
6	Pump, injector, column, detector named	1

7	Jobs of two parts	1
8	Ethanol volatile; insulin non-volatile	1
	Total	**8**

Question B5 Topic: Application of Chromatographic Methods in Pharmaceutical Analysis CH8-LO5

8 marks, C4

A regulator's lab suspects that a 'herbal' slimming capsule contains sibutramine.

(a) Explain how TLC could be used as a quick screen. (2 marks)

(b) The capsule contents are extracted and diluted 1 in 20. By HPLC, the calibration line is: peak area = 1200 + 5400 × C (C in mg/L). The diluted extract gives a peak area of 28 200. Calculate the sibutramine concentration in the diluted extract and in the original extract. (2 marks)

(c) Suggest a method to measure traces of dichloromethane left in a capsule from manufacture, with a reason. (2 marks)

(d) When measuring a drug in a patient's blood by HPLC, explain why (2 marks)

(i) the sample is cleaned up first and

(ii) an internal standard is added.

(a) Spot the capsule extract and a sibutramine reference standard side by side (and a co-spot) on a silica plate, develop and visualise (UV 254 nm or a spray). A spot in the extract with the same R_f and colour as the standard suggests sibutramine is present.

(b) C = (28 200 − 1200) ÷ 5400 = 27 000 ÷ 5400 = **5.0 mg/L** in the diluted extract; × 20 = **100 mg/L** in the original extract.

(c) Headspace **GC** (with FID or MS): dichloromethane is volatile, so it vaporises into the headspace and can be separated in the gas phase.

(d) **(i)** Blood proteins and other material would clog the column, bind to it and give interfering peaks, so they are removed (for example, by solid-phase extraction).

(ii) The internal standard is carried through every step, so it corrects for small, variable losses during extraction and injection.

Step	Marking criteria	Marks
1	Extract and standard run on the same plate	1
2	Same R_f and colour → suggests sibutramine	1
3	5.0 mg/L in diluted extract	1
4	100 mg/L in original extract	1

5	Headspace GC	1
6	Reason: volatile	1
7	Clean-up protects column, removes interferences	1
8	Internal standard corrects for losses	1
	Total	**8**

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Adapted from David Harvey, *Analytical Chemistry 2.1*, chapter 12, and *Instrumental Analysis* 28.6, LibreTexts, CC BY-NC-SA 4.0. TLC questions, drug questions and pharmacy scenarios are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.